

Dhruv Agrawal

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Education

Worcester Polytechnic Institute, Worcester, MA

August 2024 - May 2026

Master of Science in Robotics Engineering; GPA: 4.0/4.0

Visvesvaraya National Institute of Technology, Nagpur, India

July 2018 - May 2022

Bachelor of Technology in Electronics and Communication Engineering; GPA: 8.7/10.0

Relevant Courses: Robot Control, Computer Vision, Motion Planning, Navigation, Machine Learning

Skills

- **Languages:** Python, C, C++
- **Tools and Frameworks:** PyTorch, OpenCV, ROS2, Acados, PX4Autopilot, MATLAB, Docker, Git, OMPL, RViz
- **Hardware:** Crazyflie2.1+, Pixhawk, Jetson, NXP's S32K144, Syntiant's Edge AI EVB, ESP32, Arduino Uno

Experience

Robotics Intern | MVP Robotics, Bradford, VT

September 2025 – December 2025

- Performed lateral-direction stability analysis of a skid-steer mobile robot to evaluate safety under dynamic maneuvers.
- Developed a full gazebo simulation of the skid-steer mobile platform to validate the theoretical control behavior.
- Designed and tuned a PID-based torque controller for independent wheel actuation, improving low-level motion control.
- Created structured testing roadmaps and diagnosed root causes of GPS RTK position jumps through field experiments.
- Assisted the core engineering team in assembling, testing and maintenance of multiple mobile robot platforms.
- Supported automation initiatives by scripting tools to streamline base station programming and update workflows.

Firmware Engineer | Jio Platforms, Bengaluru, India

June 2022 – May 2024

- Developed **state-of-charge** and **state-of-health** estimation algorithms using **model-based** and **data-driven** methods.
- Implemented the algorithms in simulation and on **edge** devices using **TensorFlow**, while collaborating within a team.
- Created custom code for **SREC** generation in order to flash the BMS onto the microcontroller through the bootloader.
- Collaborated on the design and development of firmware for an **EV charger system** using the OCPP protocol.

Projects

Particle Filter based Drone Navigation System

- Implemented a vision-based PnP AprilTag observation model to estimate quadrotor pose from images and IMU data.
- Built a 15-state inertial navigation process model and particle filter with low variance resampling for robust pose tracking.
- Tuned measurement covariance using motion-capture ground truth and RMSE to evaluate filter accuracy and efficiency.

Kalman Filter-Based Inertial Navigation System

- Developed a non-linear strapdown INS estimating position, velocity, attitude and biases from a low-grade MEMS IMU.
- Designed and tuned a non-linear UKF using GNSS position measurements and process noise to fuse inertial and GPS data.
- Assessed filter performance against GNSS baselines using path plots and metrics including Haversine distance and RMSE.

Victoria Park Lidar SLAM System

- Built a vehicle motion model using Ackermann steering kinematics and local-level states to propagate truck pose over time.
- Formulated range-bearing models for lidar and GPS measurements to localize the vehicle and estimate tree landmarks.
- Implemented SLAM to jointly track vehicle path and tree landmarks, analyzing position error and landmark uncertainty.

Polygon CBF based obstacle avoidance for cooperative manipulation of cable-suspended payload using quadrotors

- Employed **CBF** as an obstacle avoidance method along with **non-linear MPC** for payload control in a multi-rotor system.
- Exploited the **Null-Space** of the Wrench vector to enable motion in confined spaces without affecting payload pose.
- Used **CasADi** as a problem formulation tool and **Acados** as a solver to generate control input to the MAV-payload system.
- Evaluated the effectiveness of the method while avoiding obstacles and having **safety distance constraints** on the robot.

Structure for Motion and NeRF

- Sparsely reconstructed a 3D scene while simultaneously obtaining camera poses of a monocular camera w.r.t. the scene.
- Synthesized novel views of complex scenes by optimizing the underlying continuous volumetric function from sparse inputs.

Deep and Un-Deep Visual Inertial Odometry

- Implemented a filter-based stereo visual inertial odometry that uses the **Multi-State Constraint Kalman Filter**.
- Constructed a **VIO** pipeline leveraging CNN and LSTM architectures to estimate the relative pose between two image frames.

Real Time Obstacle Avoidance & Path Planning with Kinodynamic Constraints

- Developed an **autonomous navigation system** for drones which have the ability to operate in unknown environments.
- Incorporated **kinodynamic constraints** in the path planner to improve the success rate when no constraints are added.
- Created a path re-planning method to plan sub-paths inside the visible regions and combine them to form a global path.

Quadrotor Control

- Implemented an **LQR controller** to track the trajectory of a quadrotor and compared it with a **PD controller** in simulation.
- Evaluated and analyzed the path time and distance error metrics for the LQR controller with respect to the PD controller.
- Implemented a PID controller on the Crazyflie for trajectory tracking using motion capture feedback.

Robotic Arm Manipulation

- Programmed a robotic arm in ROS2 to **grasp and pick up** simple objects whose location is available to the robot.
- Implemented **velocity kinematics** for the robot to move at a constant velocity by providing incremental position values.
- Formulated a **PD controller** to control the position of the robot joints by controlling the input current to the joint actuator.

Out of Control Planning

- Constructed the path plan of a **pendulum** and a **non-holonomic car** system with dynamic motion constraints using RG-RRT.
- Compared the average solution length and number of nodes in the solution tree between RRT, RG-RRT and KPIECE.

Publications

"Suntracker on Rocker-Bogie mechanism", *Advances in Mechanical Engineering: Select Proceedings of ICAME 2020*, 719-726.

- Engineered an **all-terrain robot** featuring a 3D-printed Rocker Bogie system stabilized with a **differential gear mechanism**.
- Mounted an adjustable solar panel on the robot and achieved **38.96%** improvement with respect to fixed mount panels.
- Achieved **15cm** step climb ability and **45°** gradient traversal success apart from smoothly maneuvering on uneven terrain.

"Improved Sign Language Recognition and Correction Using Inception Network, MediaPipe and PyEnchant", *2nd International Conference on Paradigm Shifts in Communications, Embedded Systems, Machine Learning and Signal Processing*.

- Utilized the **GoogLeNet V4** as the Neural Network for Sign Language Recognition leveraging **MediaPipe** framework.
- Integrated a **correction mechanism** using PyEnchant to improve the accuracy of the predictions by suggesting words.
- Achieved a training accuracy of **99.69%** in 47 epochs with a combined train and test dataset of 132000 images.